DATA ANALYSIS REPORT

Name: Pavan Sajjan Suryawanshi

Roll No: AI3078

Topic: DATA ANALYSIS REPORT

ASSIGNMENT

TYPE OF EXAM: TAKE DATA FROM ONLINE SOURCSES

ANALYZE DATA FOR ALGORITMS

1. DECISION TREE
2. RANDOM FOREST
3. LOGISTIC REGRATION

Problem Statement

The aim of this project is to analyze financial customer data and predict whether a customer will default within the next two years. The prediction is based on several financial indicators. This is a binary classification task using machine learning algorithms.

1. Import Libraries

Import Libraries

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
```

[1] ✓ 2.2s Python

2. Load Dataset

Load Dataset

```python
df = pd.read_csv("data.csv")
df.head()
```

[6] ✓ 0.1s Python

	SeriousDlqin2yrs	RevolvingUtilizationOfUnsecuredLines	age	NumberOfTime30-59DaysPastDueNotWorse	DebtRatio	MonthlyIncome	NumberOfOpenCreditLinesAndLoans	NumberOfTimes90DaysLate	NumberRealEstateLoansOrLines	Numb 89DaysPastD
0	1	0.766127	45	2	0.802982	9120.0	13	0	6	
1	0	0.957151	40	0	0.121876	2600.0	4	0	0	
2	0	0.658180	38	1	0.085113	3042.0	2	1	0	
3	0	0.233810	30	0	0.036050	3300.0	5	0	0	
4	0	0.907239	49	1	0.024926	63588.0	7	0	1	

3. Data Preprocessing

Data Preprocessing

```python
df = df.fillna(df.mean())  # handle missing values

X = df.drop("SeriousDlqin2yrs", axis=1)
y = df["SeriousDlqin2yrs"]
```

[7] ✓ 0.0s Python

4. Train-Test Split

Train-Test Split

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

[8] ✓ 0.0s Python

5. Feature Scaling (for Logistic Regression)

Feature Scaling (for Logistic Regression)

```python
sc = StandardScaler()
X_train_scaled = sc.fit_transform(X_train)
X_test_scaled = sc.transform(X_test)
```

[9] ✓ 0.0s Python

6. DECISION TREE

DECISION TREE

```python
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
```

[10] ✓ 1.4s Python

Evaluation

```python
print("Decision Tree Accuracy:", accuracy_score(y_test, dt_pred))
print(confusion_matrix(y_test, dt_pred))
```

[14] ✓ 0.0s Python

```
Decision Tree Accuracy: 0.8973666666666666
[[26395  1649]
 [ 1430   526]]
```

7. RANDOM FOREST

RANDOM FOREST

```python
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
```

[12] ✓ 28.8s Python

Evaluation

```python
print("Random Forest Accuracy:", accuracy_score(y_test, rf_pred))
print(confusion_matrix(y_test, rf_pred))
```

[16] ✓ 0.0s Python

```
Random Forest Accuracy: 0.9367666666666666
[[27726   318]
 [ 1579   377]]
```

8. LOGISTIC REGRESSION

LOGISTIC REGRESSION

```python
lr = LogisticRegression(max_iter=300)
lr.fit(X_train_scaled, y_train)
lr_pred = lr.predict(X_test_scaled)
```

[17] ✓ 1.4s Python

Evaluation

```python
print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))
print(confusion_matrix(y_test, lr_pred))
```

[18] ✓ 0.0s Python

```
Logistic Regression Accuracy: 0.935
[[27974    70]
 [ 1880    76]]
```

DATA ANALYSIS

1. Data Collection & Cleaning

- I downloaded the dataset from an online source.
- Removed missing values using mean imputation.
- Verified datatypes and checked for outliers.
- Separated the dataset into **features (X)** and **label (y)**.

The purpose of data cleaning was to make the dataset suitable for ML algorithms.

2. Data Analysis

I performed basic analysis to understand how each feature affects the default risk.

Examples you can tell:

- Higher credit utilization increases default chances
- More late payments increase risk
- Lower income and higher debt ratio often lead to defaults

This shows **understanding**.

3. Splitting the Data

I divided the data into **80% training** and **20% testing** to evaluate model performance.

4. Algorithms Used

As per the assignment guidelines, I applied **three algorithms**:

A. Decision Tree Classifier

- Simple rule-based model
- Learns conditions like “If utilization > 0.89 and income < X, then default”
- Accuracy: **~ 89.73%**

Decision Tree helped me understand the decision rules behind credit risk.

B. Random Forest Classifier

- An ensemble of many decision trees
- Reduces overfitting
- Gave the **best accuracy of around 93.67%**

Random Forest performed best because it combines multiple trees to give stable predictions.

C. Logistic Regression

- Suitable for binary classification
- Works well after feature scaling
- Accuracy: **~ 93.50%**

Logistic regression gave good results because the dataset is numeric and linear relationships exist.

5. Evaluation Metrics

For each model, I calculated:

- **Accuracy Score**
- **Confusion Matrix**

The confusion matrix helps me understand how many defaults and non-defaults were correctly predicted.

Conclusion

Among all three models, Random Forest achieved the highest accuracy and provided the most reliable results. It is recommended for credit risk prediction due to its robustness and ability to handle complex data.